ABHIRUP MUKHERJEE

abhirup.mukherjee@utdallas.edu mukherjeeabhirup mukherjeeabhirup.github.io mukherjeeabhirup

RESEARCH INTERESTS

I'm a Ph.D. student in Computer Science at The University of Texas at Dallas (UTD), working with Prof. Blaine Hoak as a Graduate Research Assistant on problems at the intersection of adversarial machine learning and computer security. Broadly, I'm interested in interpretable and robust AI models in security-critical applications.

EDUCATION

The University of Texas at Dallas **August 2026 - Present**
Doctor of Philosophy in Computer Science Richardson, TX

- Advisor: Prof. Blaine Hoak

The University of Texas at Dallas **August 2024 - May 2026**
Master of Science in Computer Science Richardson, TX

- Thesis Advisor: Dr. Latifur Khan
- Thesis: Activation Steering for Smart Contract Vulnerability Detection

SRM Institute of Science and Technology **June 2019 - May 2023**
Bachelor of Technology in Electronics and Communication Engineering Kattankulathur, India

PUBLICATIONS

Conference Proceedings

- **A. Mukherjee**, L. Khan, and B. Thuraisingham, "Activation Steering for Smart Contract Vulnerability Detection: A Calibration Mechanism Bounded by Pre-Training Knowledge," *IEEE International Conference on Intelligence and Security Informatics (ISI)*, Cambridge, UK, 2026. Oral presentation. [Link] [Paper] [Poster]

Thesis

- **A. Mukherjee**, "Activation Steering for Smart Contract Vulnerability Detection," *Master's thesis, The University of Texas at Dallas*, 2026. [Thesis]

PROFESSIONAL SERVICE

External Reviewer

- ACM Workshop on Artificial Intelligence and Security (AISec) - 2026
- USENIX Security Symposium - 2027

RESEARCH EXPERIENCE

Graduate Research Assistant, *The University of Texas at Dallas, Richardson, TX* **August 2026 - Present**

- Advised by Dr. Blaine Hoak; working at the intersection of adversarial machine learning and computer security.

Master's Thesis Research, *The University of Texas at Dallas* **August 2025 - May 2026**
Advisor: Dr. Latifur Khan [Project]

- Investigated activation steering for Solidity smart contract vulnerability detection, evaluating six LLMs (7B-34B parameters) on an 11-class classification task using the SmartBugs Curated dataset.
- Computed contrastive steering vectors from vulnerable/patched code pairs and applied them at inference time without fine-tuning.
- Evaluated direction-specific confidence calibration and improvements in multi-class vulnerability classification beyond few-shot prompting; this work became a first-author IEEE ISI 2026 paper.
- Replicated and evaluated VulnPatch using *QLoRA fine-tuning* of CodeLlama on multi-GPU HPC clusters, identifying failure modes including mode collapse on small datasets and insufficient Solidity competence in the base model.

Undergraduate Researcher, *SRM Institute of Science and Technology* **May 2022 - October 2022**
Advisor: Dr. T. Rama Rao [Project]

- Led development of a cost-effective *3D reconstruction* pipeline from single 2D images for SLAM and VR applications in a three-member research team.
- Implemented and benchmarked feature matching algorithms (*SIFT*, *ORB*) and *epipolar geometry* calculations using *OpenCV*, *PyTorch*, and *Blender*.

- Optimized projection matrices to improve 3D reconstruction accuracy, surpassing traditional optimization baselines.
- Developed device-agnostic image processing techniques for rapid generation of 3D assets suitable for *autonomous navigation systems*.

WORK HISTORY

HPC Student Assistant, *The University of Texas at Dallas* **January 2025 - May 2026**

- Accelerated scientific applications such as *Siesta*, achieving *10x* speedup through multi-level parallelization and batch processing using *OpenMP*, *MPI*, and *launcher*.
- Resolved *50+* support tickets from university researchers involving debugging and optimizing parallel scientific codes in *C*, *C++*, *Fortran*, and *Python*.
- Fine-tuned large language models on HPC clusters for research applications, gaining hands-on experience with *distributed training* and *GPU acceleration*.
- Assisted in identifying optimal HPC cluster configuration settings for performance boosts of up to *25%* through systematic benchmarking using custom computation kernels and industry-standard benchmarking tools.

Product Development Engineer, *Comviva Technologies* **February 2023 - July 2024**

- Developed secure payment gateway features for *Qatar National Bank's (QNB)* Digital Wallet *iOS application*, implementing efficient data structures for high-volume transaction processing.
- Architected and optimized *REST APIs* across *80+ new screens* using *Swift* and *SwiftUI*, focusing on memory-efficient data transfers for highly responsive UIs.
- Delivered production-ready *iOS apps* at a *two-week* deployment cadence, including comprehensive software testing across *three* QA gates, contributing to *10%* product unit performance improvement.
- Implemented unit testing frameworks such as *XCTest* and *Quick/Nimble*, increasing code coverage by *30%*.
- Enhanced code quality metrics by *30%* through systematic profiling, debugging, and integration of *SonarQube* static analysis in CI/CD pipelines.

Machine Learning Research Intern, *Atos Syntel* **July 2022 - September 2022**

- Optimized *CNN-based* medical imaging models for mammogram classification using *PyTorch* and *CUDA*.
- Achieved *91% accuracy* on breast cancer detection while reducing inference time by *40%* through efficient memory management and batch processing.
- Implemented an *EDA* pipeline using *NumPy* and *Pandas* for the Breast Cancer Wisconsin dataset.
- Conducted comparative analysis of deep learning (CNN) versus traditional ML approaches (Logistic Regression: *97.07%* accuracy) to identify tradeoffs between computational efficiency and accuracy in medical diagnosis applications.
- Engaged in discussions on bias mitigation in medical AI systems, particularly for datasets affecting minority populations.

SELECTED PROJECTS

Data Pattern Observatory - HackUTD 2024, *2nd Place Overall (1,100+ participants)* **November 2024**
[Code] [Devpost]

- Developed a data observability system for the *PNC Bank* challenge to visualize patterns and identify anomalies with *85%+* accuracy as data flows across an organization.
- Architected a full-stack data processing pipeline using *Python*, *MongoDB*, *Redis*, *Elasticsearch*, and *Docker*-containerized microservices with a *Streamlit* front end.

Mars Lander Simulation Game, *NASA Space Apps Challenge* **October 2021**
[Code] [Link] [Slides]

- Led ideation and development of a Mars lander simulation game in a team of five; placed 2nd locally and was selected as a global nominee among thousands of international submissions.

NodeMCU WiFi-Controlled IoT Smart Car, *SRM Institute of Science and Technology* **July 2021**
[Demo]

- Designed and implemented an *embedded system* for a remote-controlled vehicle using an *ESP8266 WiFi SoC* and *L298N* motor driver.
- Implemented direction and speed controls in *Embedded C* for a high-power motor driver (5-35V, 25W).
- Created local network infrastructure and an *Android* application interface using *Kotlin* for wireless control.

SKILLS

Machine Learning and Cybersecurity	Machine Learning, Deep Learning, Computer Vision, Natural Language Processing, Model Interpretability, Activation Steering, Cybersecurity, Smart Contract Vulnerability Analysis, Secure Software Development
Deep Learning Tools/Frameworks	PyTorch, TensorFlow, Hugging Face Transformers, PyTorch Lightning, Scikit-Learn, Keras, OpenCV, Spark MLlib
Programming Languages	Python, C, C++, Swift, SwiftUI, Fortran, SQL, Bash, LaTeX, MATLAB
HPC and Parallel Computing	NVIDIA CUDA, OpenMP, MPI, SLURM, CuPy, High Throughput Computing
Big Data and Cloud	Apache Spark, Kafka, MongoDB, Redis, Elasticsearch, Docker, AWS (EC2, S3)
Development Tools	Git/GitHub, GitLab, Docker, CI/CD (Jenkins), Postman, SonarQube, GDB, Perf, JIRA
Operating Systems	Linux/Unix, Parallel Programming, POSIX Threads, Shell Scripting

RELEVANT COURSES

Machine Learning • Deep Learning • Computer Vision • Natural Language Processing • Big Data Management and Analytics • Database Design • Design and Analysis of Computer Algorithms • Artificial Intelligence • Operating Systems Concepts • Scientific Computing with Python • Computer Communication Networks • ARM Based Embedded System Design • Microprocessors and Microcontrollers • Digital Signal Processing • Machine Perception with Cognition

AWARDS AND ACHIEVEMENTS

- **HackUTD 2024 - 2nd Place Overall** - Achieved 2nd place among 1,100+ participants for an enterprise-grade data observability system.
- **NASA Space Apps Challenge 2021 - 2nd Place Locally and Global Nominee** - Led ideation and development of a Mars lander simulation game in a team of five, selected as a global nominee among thousands of international submissions.
- **Departmental Scholarship, SRM Institute** - Awarded a merit-based scholarship in recognition of outstanding academic performance (9.46/10 CGPA) during academic year 2020-21.

LEADERSHIP EXPERIENCE

Committee Head, Team Envision, *SRM Institute of Science and Technology* **July 2021 - December 2021**

- Recruited and managed a team of *15* members for the Electronics and Mechanical domain of the university's flagship technical festival.
- Maintained technical and financial documentation for operations of two technical campus clubs.
- Coordinated with corporate sponsors and organized technical workshops and competitions for *500+* participants.

PROFESSIONAL DEVELOPMENT AND CERTIFICATIONS

- Applied Data Science with Python Specialization (IBM/Coursera)
- Deep Learning Applications for Computer Vision (University of Colorado Boulder/Coursera)
- Python for Data Science, AI and Development (IBM/Coursera)